Lesson Planning

Complete Lesson Plan Book

Class: Class 9

Subject: Biology

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

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Unit 1: Photosynthesis: How Leaves Make Food

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Unit 1: Photosynthesis: How Leaves Make Food

Learning Objectives

• Students will write the word equation and the chemical equation for photosynthesis.

• Students will explain the roles of chlorophyll, sunlight, water and carbon dioxide.

• Students will describe how leaf structure suits it for photosynthesis.

Key Concepts

• Photosynthesis is the process by which green plants make glucose from carbon dioxide and water, using sunlight energy.

• The word equation is: carbon dioxide + water, in sunlight and chlorophyll, gives glucose + oxygen.

• Chlorophyll in chloroplasts traps sunlight energy; without it, the plant cannot make food.

• Leaves are thin and flat to catch maximum light, and stomata let carbon dioxide in and oxygen out.

• Glucose made in photosynthesis is used for energy, growth, and storage as starch.

Teaching Methodology (Step by Step)

1. Ask why a plant kept in a dark cupboard turns yellow and weak, and collect students' ideas.

2. Draw a leaf cross-section and label the chloroplasts, stomata and veins.

3. Build the word equation on the board, adding each ingredient as students name it.

4. Explain why oxygen is a by-product and where the glucose goes.

5. Students test a variegated leaf idea in pairs: which parts of the leaf would contain starch, and why?

Classroom Activities

• Equation builder: groups arrange word cards into the correct photosynthesis equation.

• Leaf detective: students label a leaf diagram and match each part to its job.

• Starch prediction: students predict which leaves will test positive for starch after a day in sunlight.

Assessment Questions

1. Write the word equation for photosynthesis.

2. Why are leaves thin and flat?

3. What happens to the glucose a plant makes during photosynthesis?

Homework

• Draw and label a leaf cross-section showing where photosynthesis happens.

• Explain in four lines why a plant dies if it gets no sunlight for weeks.

Time Allocation

40 minutes